

BILAL ARSHAD

Senior Backend & Applied AI Engineer | Technical Lead

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PROFESSIONAL PROFILE

Senior Backend & Applied AI Engineer with 13+ years in software engineering, with deep experience building backend platforms, cloud-native systems, enterprise integrations, and distributed architectures across .NET, Azure, and AWS. Combines production backend engineering with agentic AI, RAG, evaluation, and AI governance, delivered as reliable, enterprise-grade systems.

CORE EXPERTISE

Backend	C#, .NET, ASP.NET Core, Python, FastAPI, REST APIs
Applied AI	Agentic Systems, RAG, LangGraph, MCP, LLM Evaluation, Human-in-the-Loop
Cloud	Azure, AWS, Serverless, API Management, Event-Driven Systems
Architecture	Distributed Systems, Microservices, Clean Architecture, Enterprise Integration
Delivery	Terraform, Docker, Kubernetes, CI/CD, Observability, Technical Leadership

EXPERIENCE

Applied AI Engineer & Senior Backend Consultant

May 2022 – Present

Hexaware Technologies (formerly Mobiquity), Amsterdam

Consult on backend architecture and delivery for enterprise clients across banking, telecom, insurance, utilities, and financial services, owning technical decisions from solution design through production on .NET, Azure, and AWS.

Deliver enterprise backend and cloud-native platforms end to end, including secure client-access portals, event-driven microservices, enterprise API integrations, and analytics data pipelines for compliance-heavy environments.

Introduced an AI-assisted SDLC and agentic workflow practice that delivery teams now use for requirement analysis, architecture, code review, and documentation, backed by a governance layer that keeps AI-assisted delivery auditable.

Established evaluation discipline for AI features across engagements, including scenario testing, structured output validation, human-review gates, and production-readiness criteria.

Support presales with solution architecture and technical demos, and mentor engineers on backend design, cloud integration, and practical AI adoption.

Senior Consultant & Senior Backend Engineer

Jul 2021 – Apr 2022

Systems Limited, Islamabad

Led migration of legacy products to .NET 5 and React under fixed consulting deadlines, owning technical direction while staying hands-on for most of the delivery.

Improved maintainability of inherited codebases through refactoring and static analysis, reducing code smells and raising maintainability scores, while driving adoption of modern engineering practices across delivery teams.

Technical Lead & Senior Software Engineer

Mar 2016 – Jul 2021

We Are Nova, Islamabad (with time in Liverpool, UK)

Progressed Senior Software Engineer (2016–2018) › Project Lead (2018–2019) › Technical Lead (2019–2021)

Provided technical leadership across **4 product teams** in fintech, hospitality, health & safety, and enterprise SaaS, owning architectural decisions, engineering standards, and delivery.

Reduced CI/CD build time 60% with Bitbucket Pipelines, then **another 40% from that new baseline** after mono-repository restructuring, and introduced one-click store deployments and release automation.

Reduced reported defects 70% after introducing code review standards, SonarLint, and SonarQube quality gates in the pipeline.

Built, mentored, and reviewed engineering teams, and trained new engineers in the UK office on .NET Core, architecture, and SOLID principles.

Software Engineer

Nov 2014 – Mar 2016

Dealers Cloud, Islamabad

Owned delivery of the dealership platform (REST APIs, SPA, Android app) and strengthened system architecture with Onion Architecture and SOLID principles.

Software Engineer

Jan 2013 – Nov 2014

Developer Tribe, Islamabad

Owned the full mobile application lifecycle and designed mobile app architectures using MVVM, SOLID, and service-oriented principles.

SELECTED ENGINEERING WORK

Velocity AI | Multi-Agent AI SaaS Platform • Hexaware

Initiated and led engineering for a platform that turns a natural-language brief into production-ready deliverables (user stories, decks, working prototypes) through an LLM-orchestrated multi-agent pipeline.

Designed a deterministic orchestration engine coordinating ~80 specialized agent roles across manifest-driven pipelines, with durable human-in-the-loop review gates and a Claude model factory spanning the Anthropic API and AWS Bedrock.

Built a custom LLM evaluation framework that raised one pipeline's scenario pass rate from **80% to 100%** across its regression suite, on a FastAPI/PostgreSQL backend provisioned on Terraform-managed AWS infrastructure.

Stack: Python, FastAPI, LangChain/LangGraph, Claude (Anthropic API + Bedrock), PostgreSQL, Next.js/React, Terraform, AWS

Apex | AI SDLC Governance Framework • Hexaware

Led development of Apex, a governance framework layering structured, auditable delivery controls on top of AI coding assistants across **6 platforms** (Claude Code, Cursor, VS Code, Kiro, Codex, Antigravity).

Enforces a spec-driven lifecycle with automated quality gates, guardrails, and tamper-evident audit logging, configurable from solo-dev to regulated-enterprise use.

Stack: Node.js, Multi-Agent Orchestration, Guardrails & Policy Enforcement, Audit Logging, Git Automation, CI/CD

MPG Nova | Client Access Portal • Hexaware

Delivered a secure, high-performance client access portal end to end on Azure cloud-native architecture (Functions, App Services, Logic Apps, API Management, Data Factory) for complex, compliance-heavy business workflows.

Stack: C#, .NET Core, Azure Functions, Microservices, SQL Server, EF Core, Azure DevOps

Agentic AI Chat & Voice Platform | Banking & Telecom • Tamyuz, Vodafone

Designed an end-to-end agentic AI architecture POC for banking and telecom, using a Supervisor Agent to coordinate specialist agents (Billing, Sales) across voice and digital channels.

Combined LLM-driven reasoning with RAG knowledge bases and secure backend integrations, focused on enterprise-grade scalability, controlled autonomy, and observability.

Stack: AWS Bedrock, Amazon Lex/Connect, DynamoDB, OpenSearch (RAG), React, .NET, REST APIs

Hejaz Financial Services | Islamic Finance Platform • Hexaware

Built and deployed Node.js/TypeScript serverless microservices on AWS Lambda and Step Functions, with event-driven design and state machines, for a leading Islamic finance platform.

Stack: TypeScript, Node.js, AWS Lambda, Step Functions, SQS/SNS, DynamoDB

Nova Core | We Are Nova

Built a reusable application boilerplate with predefined architecture, scaffolding, and deployment standards, automating project setup and **reducing initial setup from ~45 minutes of manual work to ~1.2 seconds** via a .NET Core templating engine.

Stack: C#, .NET Core, React, Docker, Kubernetes, Bitbucket Pipelines

Dealers Cloud | Dealers Cloud

Owned the dealership management portal end to end (REST APIs, SPA, Android app), and **achieved a 6x query performance improvement** for the platform's core database workload through system analysis and bottleneck resolution.

Stack: C#, .NET, ASP.NET MVC, MS SQL Server, T-SQL

ADDITIONAL PROJECT PORTFOLIO

- **Enterprise AI Document Analysis Integration** - AI-powered document extraction and workflow automation • .NET, Azure (Hexaware)
- **Reservoir Safety** - Reservoir safety oversight: water level, structural integrity, maintenance • .NET, Azure (Hexaware • Yorkshire Water)
- **End-to-End Data Pipeline** - Data ingestion, normalization, analytics, and reporting pipeline • Azure Data Factory, Power BI (Hexaware)
- **Ascent 360** - Customer engagement platform with personalized cross-channel communication • .NET, React, Azure (Systems Limited)
- **Qbunk** - Real-time food-ordering app with Stripe payments and a Firebase backend • .NET Core, React Native (Nova)
- **CHEQS** - Health & safety risk assessment platform replacing manual document handling • .NET Core, React (Nova)
- **PaidYou** - Bill-splitting fintech app with Open Banking integration and queue-based processing • .NET Core, RabbitMQ (Nova)
- **WorkPal** - Performance management and review platform for enterprises • .NET Core, ExtJS (Nova)
- **Threes / Dots / Squares** - Matrix-based game clones with a reusable Windows Phone game engine • C#, WPF, XAML (Developer Tribe)

EDUCATION

Bachelor of Science (BS), Computer Software Engineering • International Islamic University, Islamabad, Pakistan • 2008 – 2012